

VInTiMe Eclipse Plug-in Tutorial

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1 Introduction

VINTIME (Verifier of INtegrated TImed ModElS) [2, 11] is a suite of tools that combines high-level expressive power, unassisted property-preserving model-reduction and low-level distributed model checking power to describe and verify complex real-time system designs and their properties. VINTIME comprises the integration of the tools LAPSUS, VTS, OBSSlice and ZEUS.

LAPSUS [5, 4] translates real-time system designs based on fixed-priority scheduling into timed automata. Succinctly, it uses Worst-Case Completion Times (WCCT) and Best-Case Completion Times (BCCT) – which can be calculated using analytical techniques provided by those theories – to build an abstract and analyzable model of the system based on timed automata.

Timed automata [3] are finite automata extended with positive *real* valued clocks that record the dense time elapsed between events. Clocks can be used to guard transitions, and to force control locations to be abandoned in case an invariant does not hold.

VTS [1] is a notation for expressing real-time requirements in a visual and friendly – yet powerful – language, by means of negative scenarios. A VTS scenario is basically an annotated partial order of relevant events, denoting a (possibly infinite) set of matching time-stamped executions. VTS is meant to existentially predicate on system executions.

OBSSlice [6] is an optimization tool suited for the verification of networks of timed automata using virtual observers, the same kind of observers that VTS produces. It automatically discovers the set of modeling elements that can be safely ignored at each location of the observer by synthesizing behavioral dependence information among components. OBSSlice is fed with a network of timed automata and generates a transformed network which is equivalent to the one provided, up to branching-time observation.

ZEUS [7, 8, 10] is a distributed model checker that evolves from KRONOS. As such, it works over models expressed in terms of timed automata, being its main features a rigorous correctness proof for its distributed algorithm, and a high degree of flexibility due to its software architecture approach.

The VINTIME [11] toolset frontend was implemented as an Eclipse plugin. During this tutorial the installation process will be explained and then an example will be walked through.

2 Installation

After this section, the user will have a complete installation of VINTIME, ready to use. All the VINTIME required tools such as the ZEUS, the VTS translator, OBSSlice and the LAPSUS translator are included in the VINTIME Eclipse plug-in.

2.1 Eclipse

The VINTIME plugin requires Eclipse version 3.2 or better. It can be downloaded from [14].

2.2 Graphical Editing Framework

Before installing VINTIME, first other plugins it depends on must be installed. This can be accomplished with Eclipse's built-in Software Update manager:

1. Start Eclipse.
2. Optional: if you are behind a proxy, configure it in *Windows* → *Preferences* → *Install/Update*, in the “Proxy Settings” tab.
3. Open the *Help* → *Software Updates* → *Find and Install* menu.
4. Select *Search for new features to install* and click “Next”.
5. When prompted to select Update Sites, choose “Callisto Discovery Site”, click “Finish” and then select any mirror from the list.

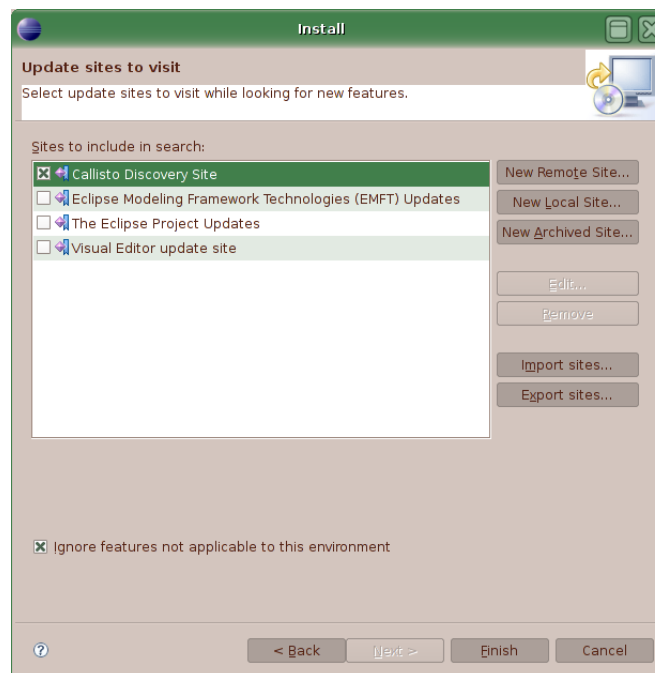


Figure 1: Callisto Discovery Site is selected

6. Select the Graphical Editing Framework (GEF) under the “Graphical Editors and Frameworks”. Accept any dependencies if needed.
7. Wait for the download to complete, review the licenses and click through any “Accept” and “Next” screens.
8. After GEF has been downloaded and installed, Eclipse must be restarted.

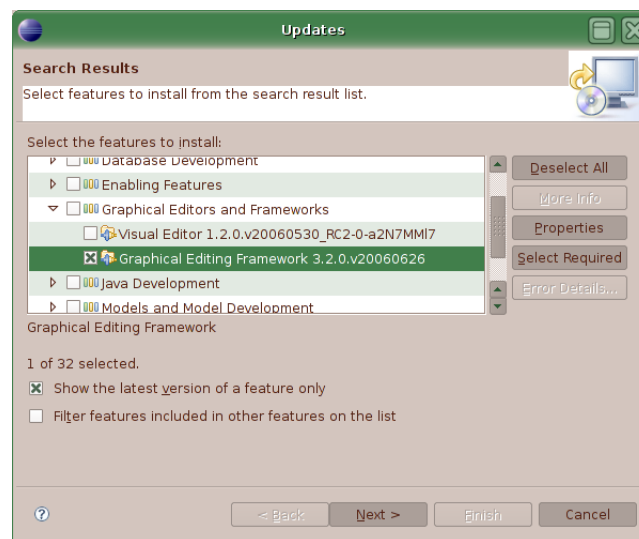


Figure 2: GEF plugin is selected

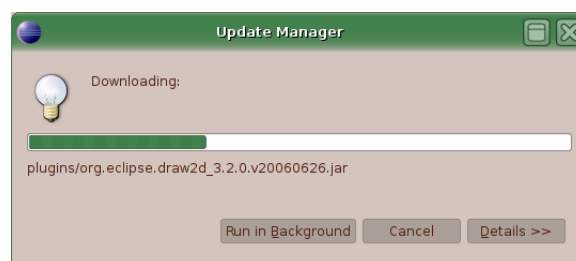


Figure 3: GEF plugin is being downloaded

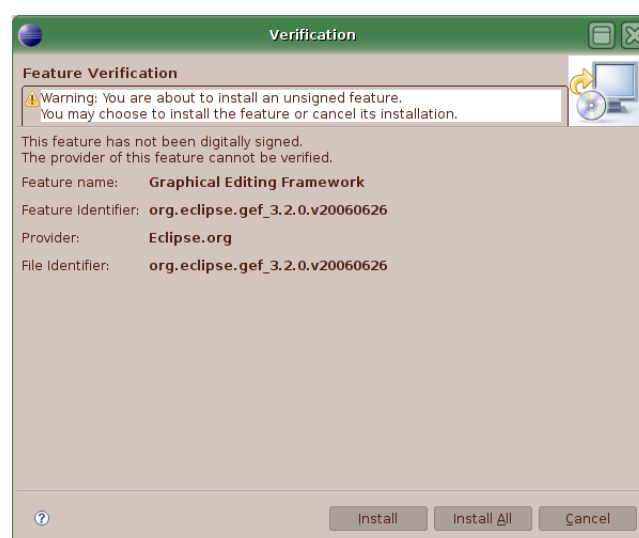


Figure 4: GEF license has to be reviewed

2.3 VInTiMe

Now the plugin can be installed. This is similar to the GEF installation.

1. Start Eclipse.
2. Open the *Help* → *Software Updates* → *Find and Install* menu.
3. Select *Search for new features to install* and click “Next”.
4. When asked to select Update Sites, click “New Remote Site”, and complete the form as follows:

Name: VInTiMe Update Site

URL: <http://dependex.dc.uba.ar/vintime/update>

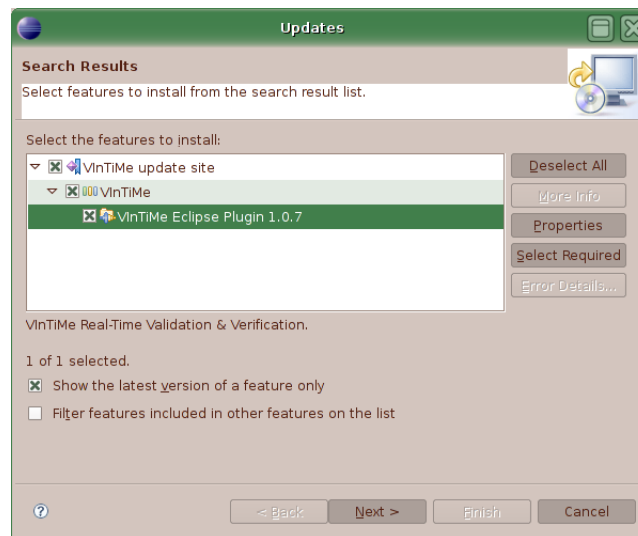


Figure 5: VInTiMe plugin is selected

5. Click *Finish* and wait for the wizard to present a list of available features to install. Select the “VInTiMe Plugin” and proceed as when installing GEF.
6. Eclipse will have to be restarted again.

2.4 Verifying the installation

To check if the installation was successful go to *Window* → *Open perspective* → *Other...* and select the “VInTiMe” perspective.

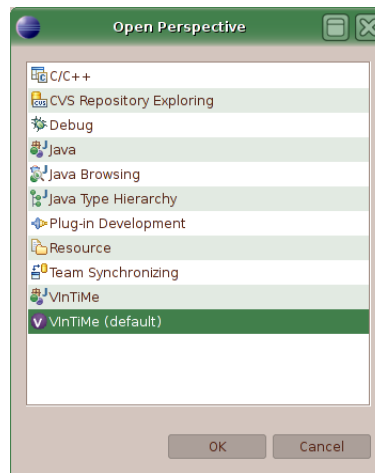


Figure 6: VInTiMe perspective is selected

If this option doesn't appear, it means that the installation was not successful. Try opening *Help* → *About Eclipse SDK* and click “Plugin details”. Check if the VINTIME plugin appears in that list.

3 Overview

VINTIME's main entity is the *Analysis*, which comprises a system under analysis (SUA) and one or more *Goals*. Goals –short for Verification Goals– determine properties for the SUA; during verification they may result in a *failure*, in which case a Counterexample *Trace* is produced. The user may inspect the resulting trace and even run a simulation on the system and produce his own traces.

The most important part of an Analysis is the SUA. A SUA is made of one or more *Components*, which are either plain *Timed Automata* (TA), *Cyclic Lapsus Tasks* or *Sporadic Lapsus Tasks*.

A designer will first create the needed Components in one editor and then group them all together within the SUA Editor. Because Components have two types of labels, local and global, integration into a single system will require the instantiation of local labels (unique names are generated in order to avoid namespace collisions) and global labels synchronization.

4 Example walkthrough

In this section we will show how VINTIME can be used to create and verify a simple example project.

We will use a subset of the Mine Pump example shown in [4]. A watchdog component emits a *ping* to a sensor periodically in order to see if it's still working properly. Another component handles the *acks* responses from the sensor and enqueues them. On each working cycle the watchdog also tries to get an *ack* from the queue.

If the sensor is working properly, then it will answer the *pings*. The ack handler will enqueue them and therefore the watchdog will always find an *ack* to extract from the queue.

But in case of sensor malfunction, the queue will eventually become empty. The watchdog will try to extract from the queue and it will fail. This will raise an alarm event.

This alarm event is received by a console proxy, which must send it through a net. On the other side, this alarm event is finally displayed.

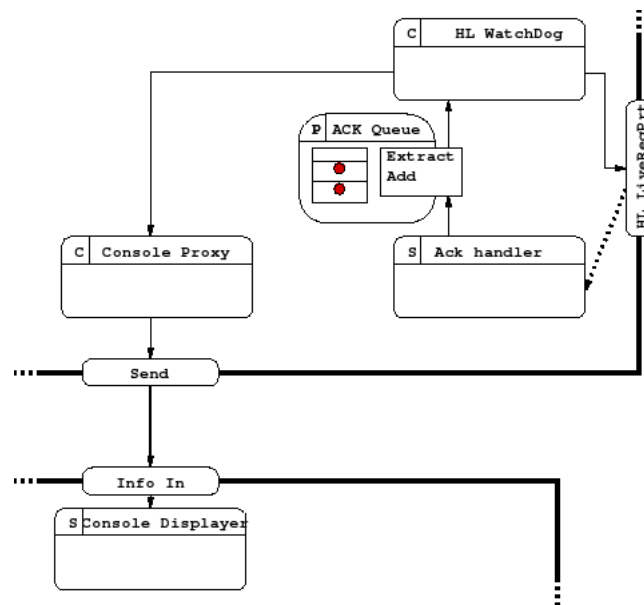


Figure 7: A part of the minepump example

We want to make sure that the alarm is displayed a short time after the sensor has actually failed.

The project used in this tutorial can be downloaded from [13] and imported into Eclipse.

4.1 Project creation

VINTIME projects can be created similarly to any other type of project: select *File* → *New* → *Project* → *VInTiMe Project*.

Create a new project and name it **Tutorial**.

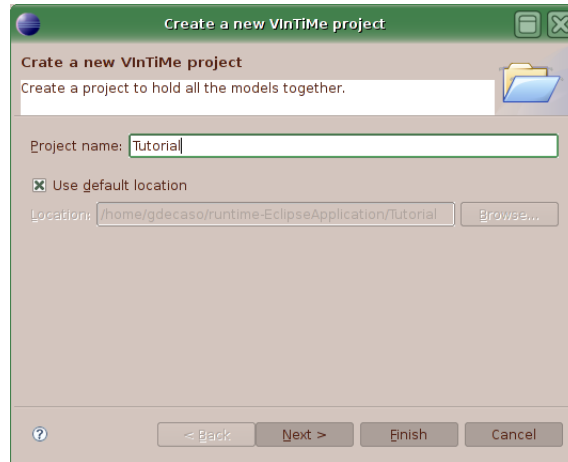


Figure 8: A new project is created

A VINTIME project consists of several folders, here is a brief explanation for each of them:

- **Analysis:** A folder for each of the defined Analysis is stored here. Each of those folders will contain the analysis and its traces.
- **Clusters:** It stores all the ZEUS clusters (sets of computers).
- **Components:** It stores the TA and LAPSUS components.
- **Goals:** It stores the VTS goals.
- **SUAs:** It stores all the Sytems Under Analysis.
- **Uppaal:** It stores the Analysis that were exported to the Uppaal format.

4.2 Components creation

As we mentioned above, the system has the following components:

- **wd**: The watchdog periodically sends a *ping* to the sensor. It also extracts from the queue and if this fails it raises an alarm. We will model it using a cyclic LAPSUS task.
- **hlw**: The hardware layer live request models the port that handles the communication from the watchdog to the sensor and from the sensor to the ack handler. We will use a TA to model it.
- **ackh**: The ack handler receives the acks from the sensor and fills the queue. We will model it using a sporadic LAPSUS task.
- **queue**: The queue stores the acks from the ack handler until the watchdog extracts them. A TA will be used to model it.
- **cproxy**: The console proxy gets the alarm event from the watchdog and sends it through the net. We will use a cyclic LAPSUS task to model it.
- **net**: The net is used to send the alarm event from the console proxy to the console displayer. The net will be modeled using a TA.
- **latch**: The latch component is used to get the alarms from the net and send them to the console displayer. This auxiliary latch component will be modeled with a TA.
- **cdisp**: The console displayer gets the alarm from the latch and displays it. The console displayer will be modeled with a sporadic LAPSUS task.

In this section we will explain how to create components in general. The finished components can be found in the tutorial project, available in [13].

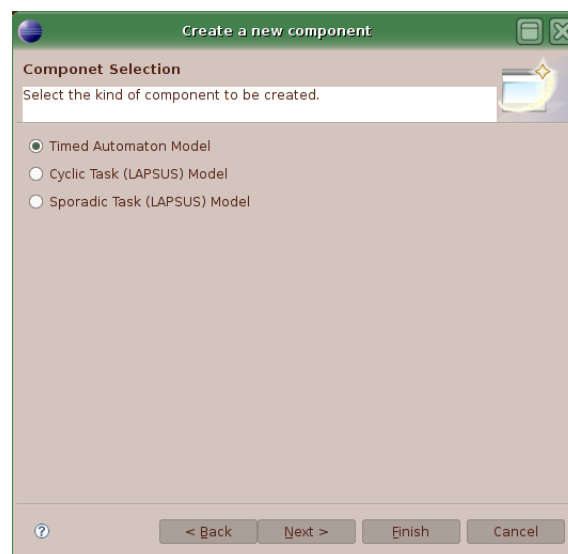


Figure 9: A new component is created

To create a component go through the following instructions:

1. Positioned in the **Tutorial** project select *File* → *New* → *Component*.
2. Select the type of component: Timed automaton, cyclic LAPSUS task or sporadic LAPSUS task. Click “Next”.
3. Enter the name for the component.
4. Click “Finish”.

Editing TA components

A timed automaton has different properties that can be edited from the “Properties View”:

- Clocks: The list of clocks that this timed automaton uses. It must be disjoint from any other clock set of the other components.
- Description: A brief description for this component.
- Name: The component name.

To add locations to a timed automaton drag them from the palette. Edges between nodes can be added using the “Transition” tool, also in the palette.

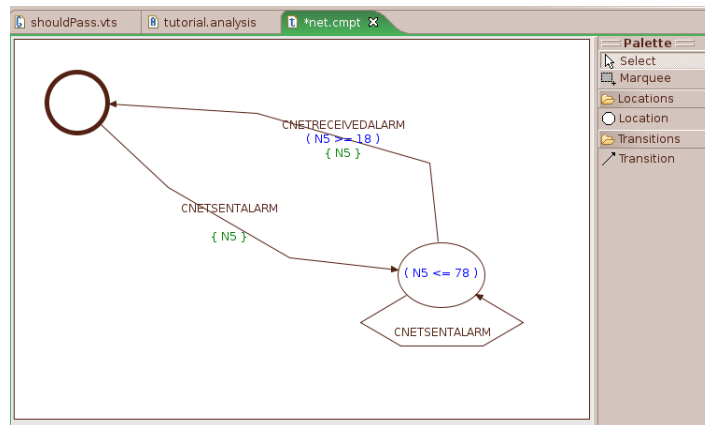


Figure 10: A TimedAutomaton being edited

Each location has different properties:

- Invariant: A clock restriction that must hold whenever this location is visited. If this invariant is not satisfied then the location must be abandoned by taking any available transition.
- Name: An optional name for the location.
- Properties (Advanced): This is internally used by the translator to mark the location as an error or trap state.

Each transition has the following properties:

- Label: The name of the event that happens when then transition is taken.
- Clocks: The set of clocks that will be reseted when the transition is taken.
- Restriction: The clock restriction that must hold in order to be able to take the transition.

Finally, the input/ouput partition can be edited from the toolbar. Look for a green and yellow button that reads “Edit input/ouput partition for labels”. From that dialog it can be defined which labels act as input sets, which of them act as output sets and which of them are local with respect to the component. It is really important to state this sets accordingly if OBSSlice will be used on the component.

Tip: On each clock, name or transition identifier, the \$ character can be used as a wildcard. When the component is instantiated in a SUA, it will be replaced by the component id.

Editing Lapsus components

Both cyclic and sporadic LAPSUS tasks share the following properties that can be edited from the “Properties View”:

- Description: A brief description of the task.
- Name: A name for the task.
- Task type (read-only): Indicates if the task is cyclic or sporadic.

Cyclic LAPSUS tasks have the following properties:

- Period: How often (in units of time) does this task execute.

Sporadic LAPSUS tasks have the following properties:

- Interruption: Tells which event label triggers the execution of this task.
- Latch: Specifies whether this task should use a latch to store interruption requests during its execution.
- MAT: The “Minimum interarrival time” specifies the minimum amount of time that passes between one interruption and the next.

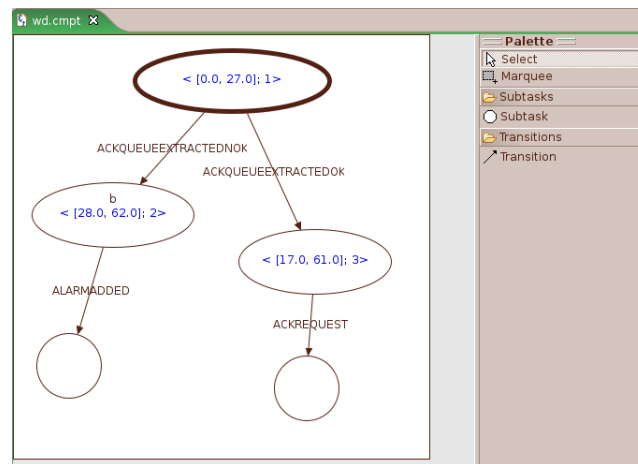


Figure 11: A Lapsus task being edited

To create subtasks and transitions between them use the palette. Remember that CDAGs can not have cycles.

To preview how this component will be translated to a timed automaton, use the “Change layout from LAPSUS to Timed Automaton” button in the toolbar (it has two yellow arrows).

Each subtask has the following properties:

- Name: An optional name for the subtask.
- Annotations: A list of annotations complying with this format $\langle I_1; P_1 \rangle - \langle I_2; P_2 \rangle - \dots - \langle I_k; P_k \rangle$. Where I_i is a time interval of the form $[d_l, d_h]$ with d_l, d_h being two floating point numbers. P_i are integer numbers stating execution priority levels (the lower the number, higher is the priority).

Bear in mind that the base priority (the priority of the first annotation of the initial subtask) must be the lowest number in the whole task. To read more about LAPSUS see [5, 4].

4.3 SUA creation

Now a SUA must be created. This is done by selecting *File* → *New* → *System Under Analysis*. Name it `tutorial.sua`.

In the Explorer at the left of your screen find the components inside the “Components” folder. Drag each of them to the SUA Editor.

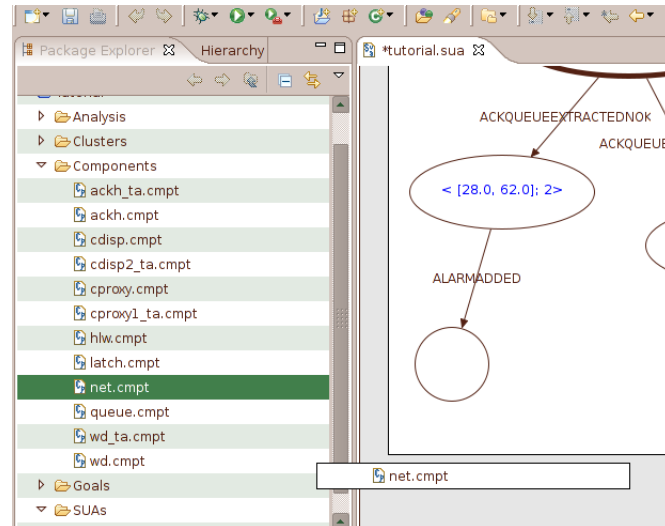


Figure 12: A component is dragged to the SUA

Once the SUA has one instance of each of the components, save it.

4.4 Goals creation

The goals for this system are available as part of the tutorial project that can be downloaded from [13].

To create the VTS Goals yourself, position in the **Tutorial** project and select *File* → *New* → *VTS Goal*.

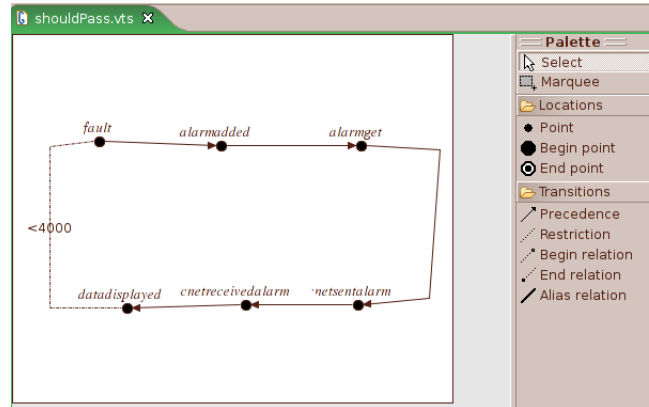


Figure 13: A VTS goal is being edited

The Goal from Figure 13 shows a series of events that begin with the failure in the sensor and end up with the alarm being displayed. The filled lines are used to state the event precedence, the dotted line in the figure creates a timed restriction between events.

The VTS editor is very rich in functionalities. To see the full syntax description you can visit [16].

4.5 Zeus cluster creation

The VINTiME Eclipse plug-in is platform independent but requires access to unix based computers in order to run verifications. A ZEUS cluster must be defined. A cluster is a set of computers that work together to solve a particular problem.

Begin by selecting *File* → *New* → *Cluster* and if asked choose a VINTiME project in which to store it.

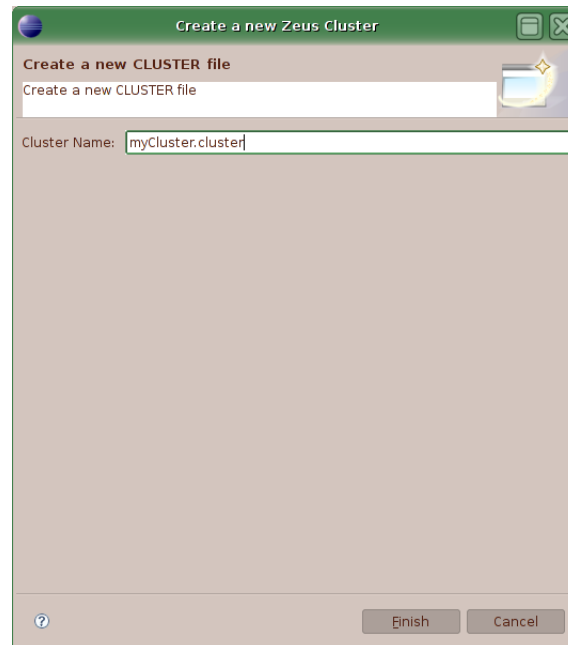


Figure 14: A new cluster is created

Add computers from the palette. You will need to add at least one computer to the cluster.

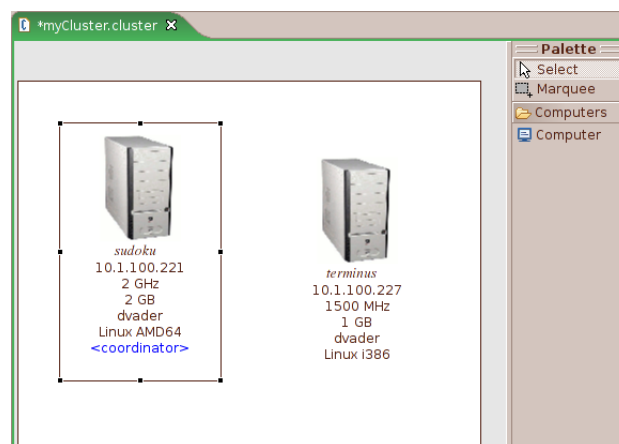


Figure 15: A cluster being edited

Edit the properties for each one and complete accurately and thoroughly the information regarding IP address, architecture, user name¹, amount of memory and clock speed. Optionally a name for each computer can be specified. One of the computers will be marked as a cluster coordinator, one and only one computer can be selected to be the coordinator by right clicking on it and selecting “Set as coordinator computer for this cluster”.

¹As used in an SSH communication to that computer.

Tip: If you want to use your computer as a node in the cluster use “127.0.0.1” as IP address.

The communication between the computer running Eclipse and the computers in the cluster is established using SSH. So before running any verification, the path to the `scp` and `ssh` commands must be specified. Go to *Window* → *Preferences...* and select the “VInTiMe preferences” category. Complete the paths to your OpenSSH compatible tools (they can be obtained from [15]).

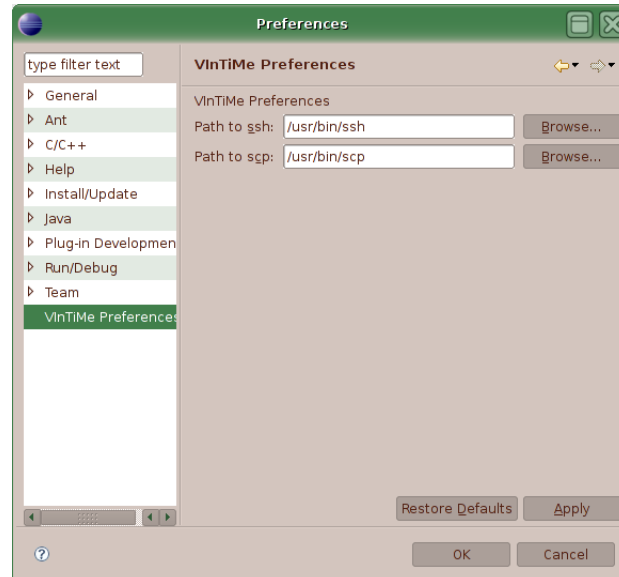


Figure 16: The VINTIME preferences screen

SSH offers the possibility to avoid entering a password when connecting to a computer. In order to use ZEUS every computer in the cluster must be reachable from the computer running Eclipse with no password. Following the example in Figure 15, a user from the computer running Eclipse should be able to connect to both “dvader@10.1.100.221” and “dvader@10.1.100.227” without having to enter a password and without having to answer any interactive question before getting a shell.

To establish a no-password connection between the “local machine” (the computer running Eclipse) and “dvader@10.1.10..221” follow these steps:

1. In the local machine run `ssh-keygen -t dsa` and press enter each time a question is asked.
2. In the local machine run `scp .ssh/id_dsa.pub dvader@10.1.100.221:~`. You will be asked for a password.
3. In the local machine run `ssh dvader@10.1.100.221`. You will be asked for a password.
4. In the remote machine make sure there is a `.ssh` directory. If not, create it running `mkdir .ssh`.
5. In the remote machine run `cat id_dsa.pub >> .ssh/authorized_keys`.
6. In the remote machine run `rm id_dsa.pub`.
7. Log out from the remote machine running `exit`.
8. Try to repeat step 3. This time you shouldn't be asked for a password.

This process has to be repeated for each of the computers in the cluster, including the coordinator.

4.6 Create an Analysis

An Analysis is the sum of a SUA and the Goals that it must satisfy. To create an Analysis go to *File* → *New* → *Analysis* and when asked which SUA to use select `tutorial.sua` from the list. Click “Next” and then select `tutorial.analysis` as a name for this Analysis.

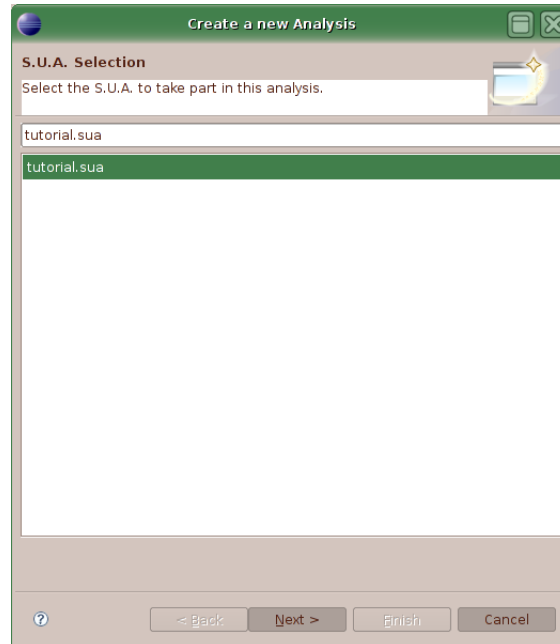


Figure 17: An Analysis being created

As was said before, the Analysis is the main entity of the plugin. From the Analysis Editor you will be able to launch ZEUS verifications, traverse traces and simulate systems to create traces from scratch.

Drag `shouldPass.vts` and `shouldNotPass.vts` from the Explorer to the Analysis Editor.

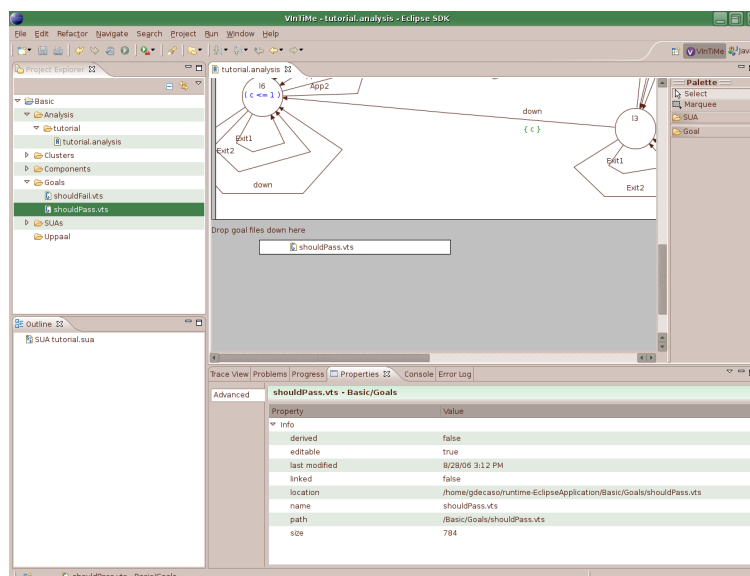


Figure 18: Drag a Goal from the explorer to the Analysis

A name and a description to both Goals can optionally be assigned. The Analysis is now completely defined, so it can be saved.

4.7 Goals verification

To run ZEUS open `tutorial.analysis`, right click on a Goal instance and select “Verify goal for this system”.

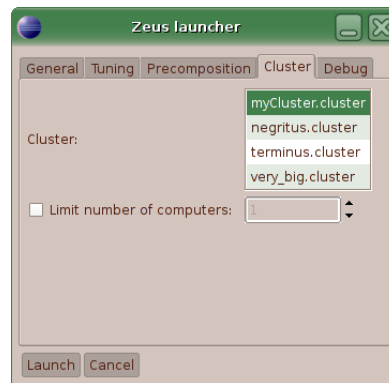


Figure 19: Zeus launch window

The ZEUS launch window will appear and from there different ZEUS options can be selected:

- General tab:
 - Generate statistics: Generates additional ZEUS information after the verification process has finished.
 - Use ObsSlice: ObsSlice reduces the verification running time by ignoring the parts of the system that do not affect the validity of a Goal.
- Tuning tab:
 - Message threshold: How many messages to wait before establishing a communication between two nodes.
 - Message timeout: How many seconds to wait before establishing a communication between two nodes.
 - Eliminate inactive clocks: The well known inactive clock elimination optimization [12].
 - Partitioning type: How to initially divide the work between the nodes.
 - Reorder clocks according to activity: The clock reordering optimization [9].
- Precomposition tab:
 - Precomposition locations limit: Limit of locations in the precomposition stage before starting the real verification.
 - Precomposition timeout: Limit of seconds in the precomposition stage before starting the real verification.
- Cluster tab:
 - Cluster: Select the cluster where the verification process will be launched. Make sure that no-password login for each of the computers in the selected cluster has already been setup.
 - Limit number of computers: Optionally a limit to the number of computers to be used during the verification can be specified here.

After the options have been selected, click on the “Launch” button. The verification process will begin and a progress bar will emerge. To continue working while the verification takes place, click on the “Run in background” button.

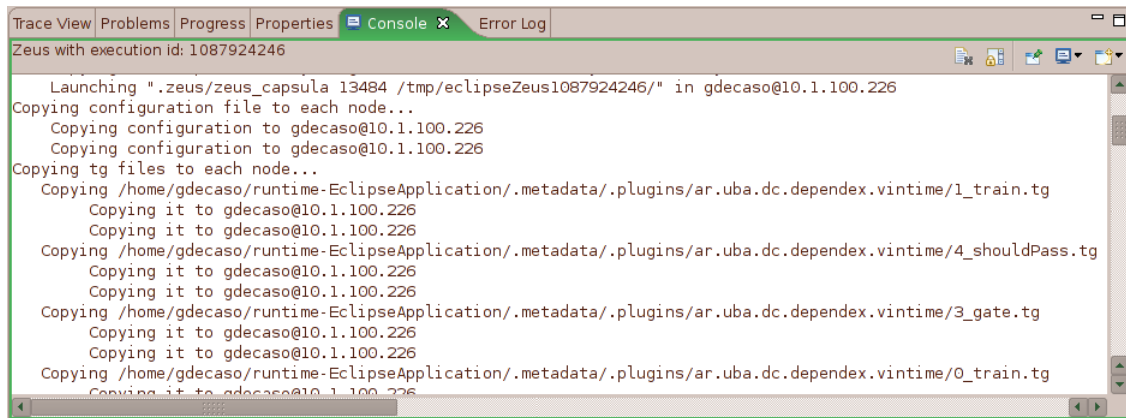


Figure 20: The Console View shows the steps of the verification process

Optionally the execution can be followed in the Console View. An explanation of the verification tasks being executed will be shown as well as the ZEUS tool output corresponding to the coordinator computer.

Once the verification has finished, a message stating if the Goal was achieved or not will emerge. Try running both of the Goals. One of them passes and the other one fails.

4.8 Trace exploring and simulation

The verification process for `shouldFail.vts` ends up returning a trace. To open it and watch the system evolve right click on the Goal instance corresponding to that Goal and select “Show trace for this goal”.

The Trace View will get focus and from there the trace can be traversed. The SUA will show the state of the system and will colour the transitions that are taken.

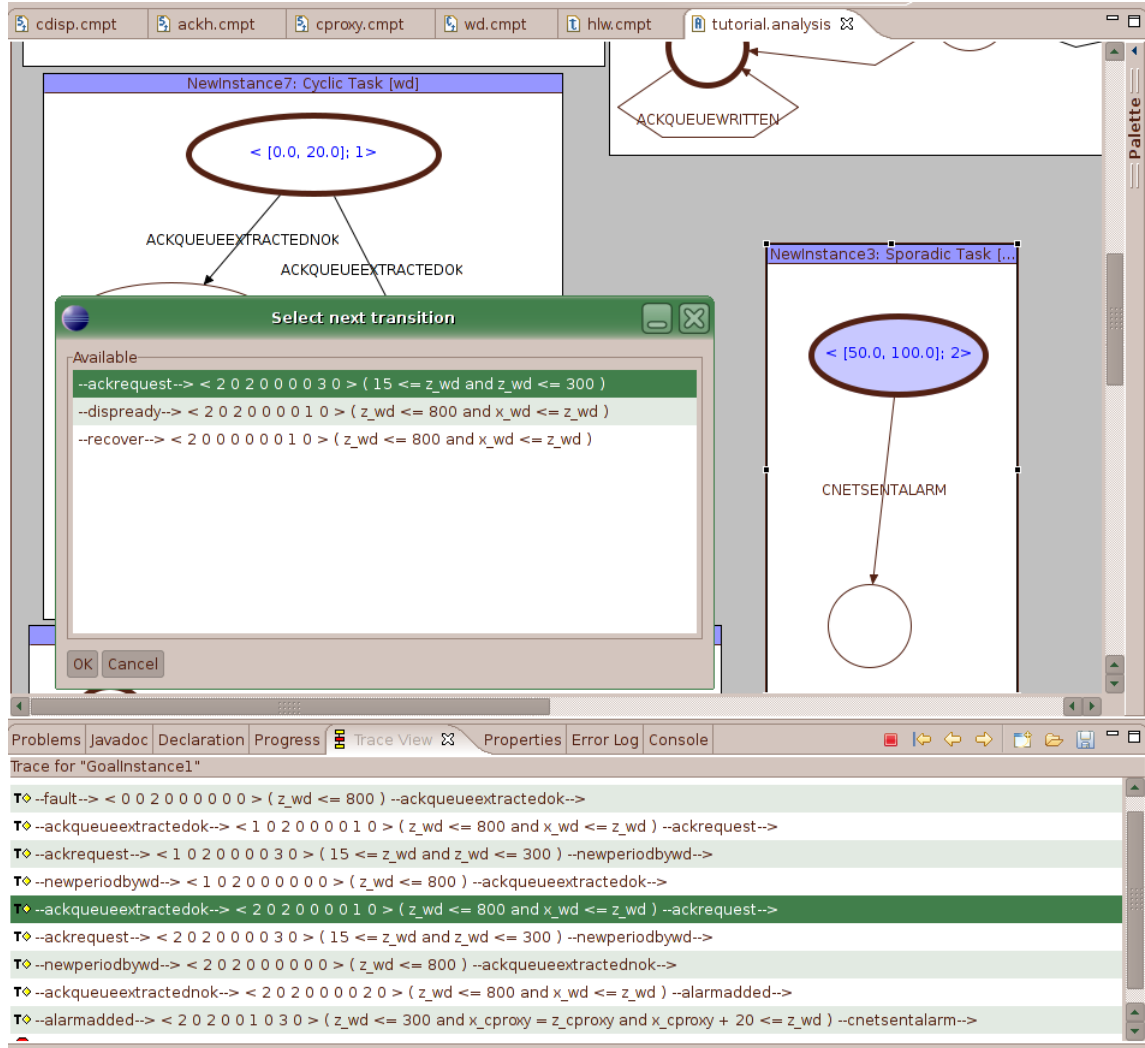


Figure 21: A trace is being extended by the user

A trace can be modified by right clicking on a step and selecting “Simulate from this point”. A set of choices that the SUA could use to evolve will be presented. Generated traces can be saved and later on loaded.

Creating a trace from scratch can be done for any Goal instance. It doesn’t matter if it has passed the verification process or not. It can even be done for Goal instances that have never been tested yet.

4.9 Other common tasks

Exporting and importing projects

To export a project in ZIP format, right-click on it and select “Export...”. From there select “Archive File” in the “General” group. Select a name for the file and click “Finish”.

Once a project is stored in a ZIP file, it can later be imported by right-clicking in the Explorer and selecting “Import...” and then “Archive File” from the “General” group. Select the ZIP file and click “Finish”.

Sending traces to other person

To send a trace to someone who already has the same project defined, open the “Analysis” folder and find the subfolder corresponding to the Analysis that was used to generate the trace. There all the traces associated to that Analysis are stored.

The name format of those traces is `goalInstanceId - traceName.trace` if the trace was saved by the user; if the trace was generated by ZEUS then the file name will be `goalInstanceId.trace`.

Send the “.trace” file to the other person and tell him to store it in the subfolder corresponding to his equivalent Analysis. If the other user has a Goal with the same ID then he will be able to open the trace by right clicking on it and selecting “Open trace view for this goal”.

If the trace was generated by ZEUS it will be automatically loaded. If the trace was saved by the user (i.e., has a hyphen in its name) the other user will have to click on the “Import” button on the Trave View and select it from the list of available traces for that Goal.

Sharing a project using CVS

VINTIME projects have the same properties that a regular Eclipse project. In particular it can be stored in a CVS repository using the “Team” submenu in the context menu that emerges when right-clicking on it.

Use the “Share project...” option to start sharing your project. Specify the CVS host, path, mode, username and password. Once the project is being shared, the “Commit” and “Update” commands can be found in the “Team” submenu.

A Glossary

- **Analysis:** This is the main entity for *VInTiMe*. It is composed of a *SUA* and a set of *Goals* to verify. Explained in page 17.
- **Component:** Components may be *TAs* or *Tasks*. Several Components together form a *SUA*. Explained in page 10.
- **Direct editing:** Changing a model's property (e.g., a label) by pointing and clicking at it, then typing the new value, as opposed to changing it from the *Properties* view.
- **Goal:** A negative scenario for the system. If a Goal is fulfilled, then an example *Trace* showing how to reach the Goal is generated. Explained in page 14.
- **Lapsus:** A translator that creates timed automata from real-time system designs based on fixed-priority scheduling [5, 4]. Explained in page 12.
- **ObsSlice:** An optimizer that is applied to the *SUA* before the verification is launched [6]. It reduces the time that takes to verify a property by “shutting down” the components that do not affect its validity. Explained in page 18.
- **SUA:** The System Under Analysis. Together with a set of *Goals*, this will form an *Analysis*. Explained in page 13.
- **System:** See *SUA*.
- **TA:** See *Timed Automaton*.
- **Task:** See *LAPSUS*.
- **Timed Automaton:** An automaton with a set of clocks and temporal restrictions in its transitions [3]. Explained in page 11.
- **Trace:** A trace contains a series of steps which show how to fulfill a *Goal*. Explained in pages 20 and 21.
- **VInTiMe:** Verifier of INtegrated TImed ModEls [2, 11]. The main plugin and toolset.
- **VTS:** Visual Timed Scenario [1]. See *Goal*.
- **Zeus:** A distributed real-time systems model checker [7, 8, 10]. Explained in page 18.

References

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